WORKED SOLUTION

STEP II 2015, Question 10

Mechanics • Constrained Motion

STEP II 2015, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

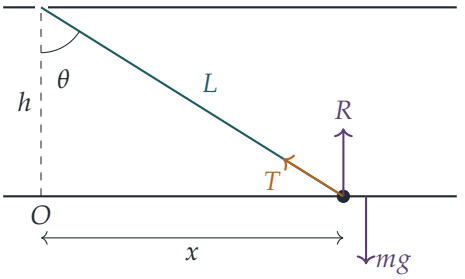
step.maths.org/questions/15-s2-q10

The key idea

This is a constraint problem: the particle's motion is completely fixed by the string, so the kinematics come first and the forces afterwards. Write the length of string below the ceiling in terms of the particle's horizontal distance from the point under the hole, and differentiate with respect to time; the rate of change of that length is known. Once the acceleration is known, Newton's second law horizontally gives the tension, and resolving vertically gives the normal reaction, which must stay non-negative while the particle is on the floor.

Solution

We model the particle as a point on a smooth horizontal floor and the string as light and inextensible, so the tension T is the same along it. Let O be the point of the floor directly below the hole, let x be the distance of the particle from O, and let L be the length of string between the hole and the particle.



The speed

By Pythagoras, and from the triangle,

$$L^2 = x^2 + h^2, \qquad x = h\tan\theta, \qquad \sin\theta = \frac{x}{L}.$$

The string is drawn in through the hole at speed V, so $\dot{L} = -V$. Differentiating $L^2 = x^2 + h^2$ with respect to time (h is constant):

$$2L\dot{L} = 2x\dot{x} \quad \Longrightarrow \quad \dot{x} = \frac{L}{x}\dot{L} = -\frac{VL}{x} = -\frac{V}{\sin\theta}.$$

So the particle moves towards O with speed

$$\boxed{v = \frac{V}{\sin\theta} = V\operatorname{cosec}\theta}$$

which is always larger than V (it is the string, not the particle, that moves at V).

Ask yourself Why isn't the particle's speed simply V, or $V\sin\theta$?

Only the component of the particle's velocity *along the string* must equal the rate at which the string is drawn in. That component is $v\sin\theta$, since the string makes angle $\frac{\pi}{2} - \theta$ with the floor. Setting $v\sin\theta = V$ gives the same answer as above, and makes clear why $v > V$.

The acceleration

Differentiate $\dot{x} = -VL/x$ again, using the quotient rule, $\dot{L} = -V$ and $\dot{x} = -VL/x$:

$$\ddot{x} = -V\,\frac{\dot{L}\,x - L\,\dot{x}}{x^2} = -V\,\frac{-Vx + VL^2/x}{x^2} = -\frac{V^2(L^2 - x^2)}{x^3} = -\frac{V^2h^2}{x^3}.$$

With $x = h\tan\theta$ this is $\ddot{x} = -\dfrac{V^2}{h\tan^3\theta}$. Since $\dot{x}$ and $\ddot{x}$ are both negative, the particle is speeding up: its acceleration has magnitude

$$\boxed{a = \frac{V^2}{h\tan^3\theta} = \frac{V^2\cos^3\theta}{h\sin^3\theta}}$$

directed towards O. (As a check, $v = V/\sin\theta$ does increase as θ decreases.)

The tension, and leaving the floor

The forces on the particle are its weight mg, the normal reaction R from the floor (vertical, as the floor is smooth) and the tension T along the string, which has horizontal component $T\sin\theta$ towards O and vertical component $T\cos\theta$ upwards.

Horizontally, by Newton's second law,

$$T\sin\theta = ma = \frac{mV^2}{h\tan^3\theta} \quad \Longrightarrow \quad \boxed{T = \frac{mV^2}{h\sin\theta\tan^3\theta} = \frac{mV^2\cos^3\theta}{h\sin^4\theta}}$$

Vertically, the particle is in equilibrium while it is on the floor, so

$$R + T\cos\theta = mg \quad \Longrightarrow \quad R = mg - \frac{mV^2\cos\theta}{h\sin\theta\tan^3\theta} = mg - \frac{mV^2}{h\tan^4\theta}.$$

Ask yourself Which way does R change as the particle is pulled in?

As the particle approaches O, θ decreases, so $\tan^4\theta$ decreases and R decreases steadily. At the start (large θ) R is close to mg; the particle stays on the floor as long as $R \geq 0$, and leaves it at the first moment that $R = 0$, which is when

$$\frac{mV^2}{h\tan^4\theta} = mg \quad \Longleftrightarrow \quad \tan^4\theta = \frac{V^2}{gh},$$

as required. For smaller θ the formula would give $R < 0$, which is impossible, so the particle has indeed left the floor.

Checks

- Dimensions: V^2/h is an acceleration, mV^2/h a force, and $V^2/(gh)$ is dimensionless, as $\tan^4\theta$ must be.
- Limits: as $\theta \to \frac{\pi}{2}$ (particle very far away) the string is almost horizontal, $v \to V$ and $a \to 0$, as expected. As $\theta \to 0$ the speed and the tension blow up, so the particle must leave the floor before reaching O.
- Numbers: with $V = 2$, $h = 1$, $\theta = \frac{\pi}{4}$ we get $v = 2\sqrt{2} \approx 2.83$ and $a = 4$; differentiating $x(t) = \sqrt{(L_0 - Vt)^2 - h^2}$ numerically gives the same values. With $g = 9.8$, the particle leaves the floor when $\theta \approx 38.6°$.

Common slips

- Taking the particle's speed to be V, or $V\sin\theta$, instead of $V/\sin\theta$.
- Forgetting that L changes when differentiating $\dot{x} = -VL/x$ a second time.
- Assuming the tension is constant because the string moves at constant speed; the particle accelerates, so it is not.

Beyond STEP

The particle here has only one degree of freedom: once the string's motion is prescribed, its position is fixed at every time. Problems with such constraints are handled systematically in Lagrangian mechanics, met in the first or second year of a university course, where the constraint is built into the choice of coordinates and constraint forces like R and T are found afterwards. Kleppner and Kolenkow, *An Introduction to Mechanics*, has many problems of this kind.

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